Necessary and Sufficient Conditions for the Existence of Ideal Linear Secret Sharing Schemes for Arbitrary Access Structures

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Abstract

Determining whether an arbitrary access structure can be realized by an ideal linear secret sharing scheme is an important research topic. we use linear codes as the main tool to construct matrices $\boldsymbol{H}$ and $\boldsymbol{G}$ over a finite field $\mathbb{F}_q$ for a given access structure $\boldsymbol{\Gamma_{\min}}$, and show that a necessary and sufficient condition for the existence of an ideal linear secret sharing scheme realizing $\boldsymbol{\Gamma_{\min}}$ is that the equation $\boldsymbol{G}\boldsymbol{H}^{\intercal} = \boldsymbol{0}$ has a solution. If this equation has a solution, then $\boldsymbol{H}$ serves as the parity-check matrix of a linear code that realizes $\boldsymbol{\Gamma_{\min}}$, and $\boldsymbol{G}$ is the corresponding generator matrix. Furthermore, we prove that the result is equivalent to the following statement: there exists an ideal linear code for realizing the $\boldsymbol{\Gamma_{\min}}$ if and only if it is the port of a matroid that is representable over $\mathbb{F}_q$.

Keywords: ideal linear secret sharing scheme, linear codes, access structure, dual access structure, matroid

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1 Introduction

Secret sharing, as an important branch of modern cryptography, distributes a secret into multiple shares among participants such that any authorized subset of participants can collaboratively recover the secret, while any unauthorized subset obtains no information about it.The schemes proposed by Shamir [1] and Blakley [2], who independently introduced the concept in 1979, laid the foundation for secret sharing theory and are both linear in nature. However, a precise formulation of linear secret sharing schemes was first provided by Beimel [3] in his 1996 Ph.D. dissertation, where a secret sharing scheme is defined to be linear if its sharing function is linear. As the schemes of Shamir and of Blakley, designed for threshold access structures, are often inadequate for handling complex practical scenarios, this limitation prompted research into secret sharing schemes for general access structures. In 1989, Ito et al. [4] constructed the first linear secret sharing scheme for general access structures, a theoretical breakthrough that spurred subsequent research [5, 6]. Monotone span programs (MSPs), a primary tool for constructing linear secret sharing schemes, were formally defined by Karchmer et al. [7], who also established the equivalence between MSPs and linear secret sharing schemes. Beimel [3] further elaborated this correspondence, and Xiao et al. [8] utilized MSPs to construct linear secret sharing schemes for general access structures.

Due to the one-to-one correspondence between linear codes and monotone span programs [9], linear codes have also become a fundamental tool for constructing linear secret sharing schemes. As early as 1981, the connection between Reed–Solomon codes and Shamir's threshold scheme was noted [10]. In 1993, Massey [11] pointed out the correspondence between minimal authorized sets and minimal codewords in the dual code whose first component is 1, and proposed secret sharing schemes based on linear codes. Building on Massey's framework, Anderson et al. [12] further investigated robust secret sharing schemes with sparse access structures and applied linear codes such as

Reed–Muller codes to secret sharing. Recently, Tang et al. [13] employed Massey's theory to construct optimal binary linear codes realizing arbitrary access structures. Meanwhile, minimal linear codes play a significant role in secret sharing, as they give rise to secret sharing schemes with determinable access structures; thus the study of minimal linear codes is of great importance. In 1998, Ashikhmin and Barg [14] investigated the properties of minimal vectors in general linear codes and established a sufficient condition for a linear code to be minimal. This condition has become an important tool in subsequent research (e.g., [15, 16]). Cohen et al.[17] observed that the sufficient condition proposed by Ashikhmin and Barg is merely sufficient but not necessary, and they further constructed counterexamples to illustrate this. Later, Chang et al. [18] constructed the first infinite family of minimal binary linear codes violating this sufficient condition. Ding et al. [19] derived necessary and sufficient conditions for a binary linear code to be minimal and constructed three infinite families of minimal binary linear codes. Furthermore, based on Ding et al.'s characterization, Li et al. [20] employed Boolean functions to construct four infinite families of minimal binary linear codes that also fail to meet the Ashikhmin–Barg condition.

Another line of research relevant to this work concerns representable matroids. Matroids, which abstract the notion of independence in combinatorial mathematics, were first defined by Whitney [21] in 1935. The concept of matroids has since provided a key theoretical foundation for characterizing ideal access structures. In 1991, Brickell and Davenport [22] established a partial characterization of ideal access structures using matroids. Their central result states that if an access structure is the port of a representable matroid, then it is ideal; conversely, if an access structure is ideal, then it is the port of a unique matroid. Following this line, Martin [23] developed an algorithm to construct the corresponding matroid from an ideal access structure. Nevertheless, a complete characterization of ideal access structures remains an open problem, and only specific families have been fully characterized, such as hierarchical threshold access structures [24], access structures defined by graphs [22, 25, 26], bipartite access structures [27], and access structures with three or four minimal authorized sets [28]. These characterizations share a common feature: all such access structures are realized by ideal linear secret sharing schemes. Although, in theory, an ideal access structure does not necessarily imply the existence of a linear secret sharing scheme that realizes it, linear schemes are more efficient in practice. Therefore, necessary and sufficient conditions for an arbitrary access structure to be realized by an ideal linear secret sharing scheme are still of practical significance to explore.

In this direction, our work makes the following main contributions:

- **Correction of constraints in the original theorem.** Tang et al.[29] first established that the existence of an ideal linear secret sharing scheme realizing an access structure $\Gamma_{\min}$ is equivalent to the solvability of a system of quadratic congruences $GH^{\top} = 0$ over some finite field $\mathbb{F}_q$ [29, Theorem 2]. Let the collection of all maximal unauthorized sets R_i corresponding to $\Gamma_{\min}$ be denoted the adversary structure $\mathcal{R}(\Gamma_{\min})$. Tang et al. asserted that when constructing the matrix $G = (g_{ij})$ associated with $\mathcal{R}(\Gamma_{\min})$, if $j \notin R_i$, then $g_{ij} \in \mathbb{F}_q$ is an unknown that may possibly be zero.

  In Section 3, we revisit this theorem from the novel perspective of access structures and their duals, and we prove that the unknowns g_{ij} must in fact be nonzero elements of $\mathbb{F}_q$. This correction refines and completes the original theorem.

- **Revelation of the duality between matrices G and H.** During the proof of the theorem, we elucidate the dual relationship between the matrices G and H in an ideal linear secret sharing scheme that realizes the access structure, namely, that the row space of G is the orthogonal complement of the row space of H (and vice versa). This relationship was not explicitly clarified in the original work [29]. Based on this new insight, we present a more comprehensive version of the theorem (Theorem 7).

- **Bridging three perspectives on the same mathematical object.** From the matroid point of view, we uncover a profound connection among ideal linear secret sharing schemes, linear codes, and representable matroids for a given access structure. This provides an equivalent criterion that bridges cryptography, algebraic coding theory, and combinatorial mathematics in the study of the same underlying object.

The remainder of this paper is organized as follows: Section 2 reviews basic concepts and relationships among access structures, linear secret sharing schemes, linear codes, and matroids. Section 3 presents the necessary and sufficient conditions for an access structure to be realized by an ideal linear code. Section 4 discusses the equivalence among the existence of an ideal linear secret sharing scheme, the solvability of a system of quadratic congruences under a specific linear code construction, and the representability of a matroid for the same mathematical object; illustrative examples are also provided. Section 5 concludes the paper and outlines future research directions.

2 Preliminaries

2.1 Access Structures and Secret Sharing Schemes

For convenience, we denote the cardinality of a finite set X by $|X|$, the set difference of X and Y by $X - Y$, and the power set of a finite set X by $\mathcal{P}(X)$.

Definition 1 (Access structure) Let $P = \{P_1, P_2, \ldots, P_n\}$ be a set of n participants and let $\mathcal{AS} \subseteq \mathcal{P}(P)$ be a non-empty subset of $\mathcal{P}(P)$. We call $\mathcal{AS}$ an access structure on P if it satisfies the following monotonicity property:

$$\text{if } A \in \mathcal{AS}, \text{ then for every } A' \in \mathcal{P}(P) \text{ with } A \subseteq A', \text{ we have } A' \in \mathcal{AS}.$$

Any set in $\mathcal{AS}$ is called an **authorized set** on P. Moreover, an authorized set S is said to be a **minimal authorized set** on P

$$\text{if for every } S' \in \mathcal{P}(P) \text{ with } S' \subsetneq S, \text{ we have } S' \notin \mathcal{AS}.$$

The collection of all minimal authorized sets on P is defined as the **minimal access structure** $\Gamma_{\min}$. When no confusion arises, we may also refer to $\Gamma_{\min}$ simply as the access structure.

Any set in $\mathcal{P}(P)-\mathcal{AS}$ is called an **unauthorized set** on P. Furthermore, an unauthorized set R is said to be a **maximal unauthorized set**

$$\text{if for every } R' \in \mathcal{P}(P) \text{ with } R \subsetneq R', \text{ we have } R' \in \mathcal{AS}.$$

The collection of all maximal unauthorized sets on P is defined as the **adversary structure** $\mathcal{R}(\Gamma_{\min})$ corresponding to $\Gamma_{\min}$.

Remark 1 Throughout this paper, we assume that access structures are **non-redundant**, that is, each participant appears in at least one minimal authorized set.

To analyze the duality of the above structures, we introduce the dual access structure (introduced by van Dijk [9]). The **dual access structure** of $\Gamma_{\min}$ is defined as

$$\Gamma^*_{\min} = \{S^* \subseteq P \mid P - S^* \in \mathcal{R}(\Gamma_{\min})\}.$$

Correspondingly, the adversary structure associated with $\Gamma^*_{\min}$ is defined as

$$\mathcal{R}^*(\Gamma^*_{\min}) = \{R^* \subseteq P \mid P - R^* \in \Gamma_{\min}\}.$$

Definition 2 (Secret sharing scheme) Let the secret $s \in \mathbb{F}_q$, and a secret sharing scheme realizing an access structure $\Gamma_{\min}$ on the participant set P is called **perfect** if it satisfies the following two conditions:

- **Correctness:** Every minimal authorized set $A \in \mathcal{AS}$ can uniquely reconstruct the secret s from its shares s_i.
- **Perfect Security:** Any unauthorized set $B \in \mathcal{P}(P) - \mathcal{AS}$ obtains no information about the secret s.

A perfect secret sharing scheme realizing $\mathcal{AS}$ is said to be **ideal** if the share length of each participant is equal to the length of the secret. Furthermore, if the process of sharing the secret is linear, then the scheme is called **an ideal linear secret sharing scheme**.

This paper aims to investigate necessary and sufficient conditions for an arbitrary access structure to be realizable by an ideal linear secret sharing scheme.

2.2 Secret Sharing Schemes Based on Linear Codes

With linear codes serving as a fundamental tool for constructing linear secret sharing schemes, this paper employs linear codes to investigate necessary and sufficient conditions for an arbitrary access structure to be realizable by an ideal linear secret sharing scheme.

Definition 3 (Linear code) A k-dimensional subspace C of the vector space $\mathbb{F}_q^n$ is called an $[n, k]$ **linear code** if it satisfies

$$a\boldsymbol{c} + \tilde{a}\tilde{\boldsymbol{c}} \in C, \qquad \forall a, \tilde{a} \in \mathbb{F}_q,\ \forall \boldsymbol{c}, \tilde{\boldsymbol{c}} \in C,$$

where each element $\boldsymbol{c} = (c_1, c_2, \ldots, c_n) \in C$ is called a **codeword**.

Since a linear code C is a k-dimensional subspace of $\mathbb{F}_q^n$, taking k linearly independent codewords from C as row vectors yields a $k \times n$ matrix G, which is called a **generator matrix** of C.

The **dual code** of C is defined as

$$C^{\perp} = \{\boldsymbol{c}' \in \mathbb{F}_q^n \mid \forall \boldsymbol{c} \in C,\ \langle \boldsymbol{c}, \boldsymbol{c}' \rangle = 0\},$$

where $\langle \boldsymbol{c}, \boldsymbol{c}' \rangle$ denotes the Euclidean inner product. It is well known that if C is an $[n, k]$ linear code over $\mathbb{F}_q$, then $C^{\perp}$ is an $[n, n-k]$ linear code over $\mathbb{F}_q$.

Similar to C, the dual code $C^{\perp}$ also has a generator matrix. Let H be a generator matrix of $C^{\perp}$; then H is also called a **parity-check matrix** of C.

Let C be an $[n, k]$ linear code over $\mathbb{F}_q$ and let $\boldsymbol{c} = (c_1, c_2, \ldots, c_n) \in C$ be a codeword. The **support** of $\boldsymbol{c}$ is defined as $\mathrm{supp}(\boldsymbol{c}) = \{i \in [1, n] \mid c_i \neq 0\}$. A codeword $\boldsymbol{c}$ is said to **cover** another codeword $\boldsymbol{v}$ if $\mathrm{supp}(\boldsymbol{v}) \subseteq \mathrm{supp}(\boldsymbol{c})$.

A nonzero codeword $\boldsymbol{c} \in C$ is called a **minimal codeword whose first component is 1** if it satisfies the following two conditions:

- the first component of $\boldsymbol{c}$ is 1;
- it covers no other codeword whose first component is 1, apart from its own scalar multiples.

Now let $P = \{P_1, \ldots, P_n\}$ be a set of n participants and let $\Gamma_{\min} = \{S_1, \ldots, S_m\}$ be the access structure consisting of all minimal authorized sets on P. Suppose there exists an $[n+1, k]$ linear code C over $\mathbb{F}_q$ whose parity-check matrix H is constructed as follows:

$$H = \begin{pmatrix} 1 & h_{11} & h_{12} & \cdots & h_{1n} \\ 1 & h_{21} & h_{22} & \cdots & h_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & h_{m1} & h_{m2} & \cdots & h_{mn} \end{pmatrix},$$

where $h_{ij} \neq 0$ if and only if $P_j \in S_i$. Each row of H corresponds to a minimal authorized set $S_i \in \Gamma_{\min}$. The **labeling map** ρ is defined as

$$\rho = \begin{pmatrix} 2 & 3 & \cdots & n+1 \\ P_1 & P_2 & \cdots & P_n \end{pmatrix},$$

where the first row lists the column indices of H and the second row indicates the corresponding participants.

Following Massey [11], we introduce the notion of an **ideal linear code** as defined below.

Definition 4 (Ideal linear code) For the given $\Gamma_{\min}$, we say that C is an **ideal linear code** realizing $\Gamma_{\min}$ if and only if there exists a set of nonzero elements $h_{ij} \in \mathbb{F}_q$ that yields the parity-check matrix H as above such that each row vector of H corresponds bijectively to a minimal codeword with first component 1 in the dual code $C^{\perp}$.

Such an ideal linear code directly induces a secret sharing scheme realizing $\Gamma_{\min}$. Let $G = (\boldsymbol{g}_0, \boldsymbol{g}_1, \ldots, \boldsymbol{g}_n)$ be a $k \times (n+1)$ generator matrix of C, where each $\boldsymbol{g}_i$ is a column vector over $\mathbb{F}_q$.

Secret Distribution. The dealer P_0 first randomly selects a vector $\boldsymbol{u} = (u_1, u_2, \ldots, u_k)$ with $u_i \in \mathbb{F}_q$ satisfying $\boldsymbol{u}\boldsymbol{g}_0 = s$, computes $\boldsymbol{u}G = (\boldsymbol{u}\boldsymbol{g}_0, \boldsymbol{u}\boldsymbol{g}_1, \ldots, \boldsymbol{u}\boldsymbol{g}_n)$ and then distributes the share $s_i = \boldsymbol{u}\boldsymbol{g}_i$ to participant P_i.

Secret Reconstruction. Let $S = \{P_{i_1}, P_{i_2}, \ldots, P_{i_m}\} \subseteq P$ be a minimal authorized set. By construction, the vector

$$\boldsymbol{v} = (1, 0, \ldots, 0, h_{i_1}, 0, \ldots, 0, h_{i_m}, 0, \ldots, 0)$$

is a minimal codeword with first component 1 in the dual code $C^{\perp}$. Consequently,

$$\boldsymbol{g}_0 = -\sum_{j=1}^{m} h_{i_j} \boldsymbol{g}_{i_j},$$

and thus

$$s = \boldsymbol{u}\boldsymbol{g}_0 = -\boldsymbol{u}\sum_{j=1}^{m} h_{i_j} \boldsymbol{g}_{i_j} = -\sum_{j=1}^{m} h_{i_j} (\boldsymbol{u}\boldsymbol{g}_{i_j}) = -\sum_{j=1}^{m} h_{i_j} s_{i_j}.$$

The access structure realized by this scheme is exactly $\Gamma_{\min}$, and the scheme is linear. Moreover, by the labeling map, the share length of each participant is equal to the secret length, hence the scheme is also ideal. Therefore, the existence of an ideal linear secret sharing scheme for an arbitrary access structure $\Gamma_{\min}$ is equivalent to the existence of an ideal linear code realizing $\Gamma_{\min}$ as defined above.

2.3 Matroids and Linear Codes

This subsection aims to introduce the fundamental correspondence between matroids and linear codes, which provides the necessary theoretical foundation for subsequent discussions. Based on [30], we first present basic knowledge and properties of matroids, and then introduce the correspondence between matroids and linear codes as elaborated in [31].

Definition 5 (Matroid) A matroid $\mathcal{M}$ is an ordered pair $(E, \mathcal{I})$, where E is a finite set and $\mathcal{I} \subseteq \mathcal{P}(E)$ is the family of **independent sets** of $\mathcal{M}$, satisfying the following axioms:

1. $\emptyset \in \mathcal{I}$.
2. If $I \in \mathcal{I}$ and $I' \subseteq I$, then $I' \in \mathcal{I}$.
3. If $I_1, I_2 \in \mathcal{I}$ and $|I_1| < |I_2|$, then there exists $e \in I_2 - I_1$ such that $I_1 \cup \{e\} \in \mathcal{I}$.

The members of $\mathcal{I}$ are called **independent sets** of $\mathcal{M}$. A **maximal independent set** in $\mathcal{M}$ is called a **basis** of $\mathcal{M}$, denoted by B. If $X \notin \mathcal{I}$, then X is called a **dependent set** of $\mathcal{M}$. A minimal dependent set of $\mathcal{M}$ is called a **circuit** of $\mathcal{M}$; that is, a dependent set C is a circuit if and only if every proper subset of C is independent. Furthermore, the **rank function** $r_{\mathcal{M}} : \mathcal{P}(E) \to \mathbb{N}$ of $\mathcal{M}$ is defined as the cardinality of a maximal independent set contained in $X \subseteq E$, i.e.,

$$r_{\mathcal{M}}(X) = \max\{|I| : I \subseteq X,\ I \in \mathcal{I}\}.$$

If $r_{\mathcal{M}}(E) = r$, then $\mathcal{M}$ is called a matroid of rank r.

Definition 6 (Representable matroid) Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid of rank r. $\mathcal{M}$ is said to be **representable over a field** $\mathbb{F}$ if there exists a mapping $\phi : E \to V$, where V is some r-dimensional vector space over $\mathbb{F}$, such that for every subset $X \subseteq E$, the rank of X in $\mathcal{M}$ equals the dimension of the subspace spanned by $\phi(X)$ in V, i.e.,

$$r_{\mathcal{M}}(X) = \dim_{\mathbb{F}} \langle \phi(X) \rangle.$$

Such a mapping ϕ is called an $\mathbb{F}$-**coordinatization** of $\mathcal{M}$.

Definition 7 (Connected matroid) Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid. If for any two distinct elements $x, y \in E$, there exists a circuit $C \in \mathcal{C}(M)$ such that $\{x, y\} \subseteq C$, where $\mathcal{C}(M)$ is the family of circuits of $\mathcal{M}$, then $\mathcal{M}$ is called a **connected matroid**.

Definition 8 (Dual matroid) Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and let $\mathcal{B}(\mathcal{M})$ denote the family of bases of $\mathcal{M}$. The **dual matroid** $\mathcal{M}^* = (E, \mathcal{I}^*)$ of $\mathcal{M}$ is defined by its family of independent sets $\mathcal{I}^*$ as follows:

$$\mathcal{I}^* = \{I^* \subseteq E \mid \exists B \in \mathcal{B}(M) \text{ such that } B \subseteq E - I^*\}.$$

That is, a subset $I^* \subseteq E$ is independent in $\mathcal{M}^*$ if and only if its complement $E - I^*$ contains a basis of $\mathcal{M}$. From the definition of the dual matroid, the following properties hold:

- $(\mathcal{M}^*)^* = \mathcal{M}$.
- For $X \subseteq E$, $r_{\mathcal{M}^*}(X) = |X| - r_{\mathcal{M}}(E) + r_{\mathcal{M}}(E - X)$. In particular, $r_{\mathcal{M}^*}(E) = |E| - r_{\mathcal{M}}(E)$.
- $\mathcal{M}^*$ is representable over a field $\mathbb{F}$ if and only if $\mathcal{M}$ is representable over $\mathbb{F}$.
- $\mathcal{M}^*$ is connected if and only if $\mathcal{M}$ is connected.

Definition 9 (Matroid isomorphism) Let $\mathcal{M}_1 = (E_1, \mathcal{I}_1)$ and $\mathcal{M}_2 = (E_2, \mathcal{I}_2)$ be two matroids. If there exists a bijection $\phi : E_1 \to E_2$ such that for every $X \subseteq E_1$,

$$X \in \mathcal{I}_1 \iff \phi(X) \in \mathcal{I}_2,$$

then ϕ is called an **isomorphism** from $\mathcal{M}_1$ to $\mathcal{M}_2$, and $\mathcal{M}_1$ and $\mathcal{M}_2$ are said to be **isomorphic**, denoted by $\mathcal{M}_1 \cong \mathcal{M}_2$. Intuitively, the independence structures of $\mathcal{M}_1$ and $\mathcal{M}_2$ are essentially the same.

Let $G = (\boldsymbol{g}_0, \boldsymbol{g}_1, \ldots, \boldsymbol{g}_n)$ be a $k \times (n+1)$ matrix over $\mathbb{F}_q$ that serves as a generator matrix of an $[n+1, k]$ linear code C. The column vectors of G define an $\mathbb{F}_q$-representable matroid $\mathcal{M}$ on the point set $P = \{P_0, P_1, \ldots, P_n\}$, whose rank function $r_{\mathcal{M}} : \mathcal{P}(P) \to \mathbb{N}$ is given by

$$r_{\mathcal{M}}(A) = \mathrm{rank}_{\mathrm{LIN}}\left(\{\boldsymbol{g}_i \mid i \in A\}\right), \qquad A \subseteq P,$$

where $\mathrm{rank}_{\mathrm{LIN}}$ denotes the linear algebraic rank of the set of vectors (i.e., the dimension of the subspace they span).

It is worth noting that the matroid $\mathcal{M}$ depends only on the linear code C, not on the particular choice of the generator matrix G. We say that $\mathcal{M}$ is the **matroid associated with** C, and that C is an $\mathbb{F}_q$-representation of $\mathcal{M}$. It is easy to verify that the rank of the matroid equals the rank of the generator matrix, i.e., $r_{\mathcal{M}}(P) = \text{rank}_{\text{LIN}}(G)$.

Remark 2 Different linear codes may represent the same matroid.

Furthermore, if the secret sharing scheme based on the linear code C realizes an access structure $\Gamma_{\min}$ on P (not necessarily ideal), then the secret sharing scheme based on the dual code $C^{\perp}$ realizes the dual access structure $\Gamma^*_{\min}$, and both schemes have the same information rate.

Under this correspondence, the matroid $\mathcal{M}_1$ associated with the dual code $C^{\perp}$ is precisely the dual matroid of the matroid $\mathcal{M}_2$ associated with C. Since the dual of an $\mathbb{F}_q$-representable matroid is also $\mathbb{F}_q$-representable, the duality of access structures is naturally reflected in the duality of matroids: If $\Gamma_{\min}$ is the **port** of the matroid $\mathcal{M}_1$ at P_0 — that is, a subset $S_i \subseteq P - \{P_0\}$ is a minimal authorized set in $\Gamma_{\min}$ if and only if $S_i \cup \{P_0\}$ is a circuit of $\mathcal{M}_1$ (the notion of the port of a matroid throughout the sequel follows this definition) — then $\Gamma^*_{\min}$ is the port of the dual matroid M_2 at P_0.

3 Algebraic Characterization of Ideal Linear Realizations of Access Structures

The problem of whether an access structure can be realized by an ideal linear secret sharing scheme can, from the perspective of algebraic coding theory, be transformed into the existence problem of a linear code with a specific construction. This section aims to explore the intrinsic relationship between an access structure and its dual with respect to the existence of ideal linear code realizations, starting from the naturally symmetric pair consisting of an access structure and its dual.

3.1 From the Existence of a Scheme to an Algebraic Condition

Let $P = \{P_1, \ldots, P_n\}$ be a set of n participants and $\Gamma_{\min} = \{S_1, \ldots, S_m\}$ be an access structure defined on P. Suppose there exists an $[n+1, k_1]$ linear code C over $\mathbb{F}_q$ whose parity-check matrix H is constructed as follows:

$$H = \begin{pmatrix} 1 & h_{11} & h_{12} & \cdots & h_{1n} \\ 1 & h_{21} & h_{22} & \cdots & h_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & h_{m1} & h_{m2} & \cdots & h_{mn} \end{pmatrix},$$

where $h_{ij} \neq 0$ if and only if $P_j \in S_i$. Each row of H corresponds to a minimal authorized set $S_i \in \Gamma_{\min}$. The labeling map ρ is defined as

$$\rho = \begin{pmatrix} 2 & 3 & \cdots & n+1 \\ P_1 & P_2 & \cdots & P_n \end{pmatrix},$$

where the first row lists the column indices of H and the second row indicates the corresponding participants.

Furthermore, let $\Gamma^*_{\min} = \{S^*_1, \ldots, S^*_l\}$ be the dual access structure of $\Gamma_{\min}$. Suppose there exists an $[n+1, k_2]$ linear code D over $\mathbb{F}_q$ whose parity-check matrix G is constructed as follows:

$$G = \begin{pmatrix} 1 & g_{11} & g_{12} & \cdots & g_{1n} \\ 1 & g_{21} & g_{22} & \cdots & g_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & g_{l1} & g_{l2} & \cdots & g_{ln} \end{pmatrix},$$

where $g_{ij} \neq 0$ if and only if $P_j \in S_i^*$. Each row of G corresponds to a minimal authorized set of $\Gamma_{\min}^*$, and the labeling map is the same as above.

Based on the construction of the parity-check matrices H and G described above, the necessity for the existence of an ideal linear code realizing $\Gamma_{\min}$ can be clearly presented as follows: If $\Gamma_{\min}$ is realized by an ideal linear code with parity-check matrix H, then the matrix equation $GH^\mathsf{T} = 0$ has a solution over $\mathbb{F}_q$. This is not a coincidence; it is determined by the duality of linear codes and the duality of the access structures they realize. More specifically, we provide the following rigorous proof.

Theorem 1 *If $\Gamma_{\min}$ can be realized by an ideal linear code C with parity-check matrix H, then $GH^\mathsf{T} = 0$ has a solution over $\mathbb{F}_q$.*

Proof Assume there exists an ideal linear code C realizing $\Gamma_{\min}$. That is, there exists a set of nonzero elements $h_{ij} \in \mathbb{F}_q$ that yields the matrix H as above, such that the supports (excluding the first coordinate) of the minimal codewords with first component 1 in the row space of H are in bijective correspondence with the minimal authorized sets in $\Gamma_{\min}$. Hence C is a linear code with parity-check matrix H. Consequently, $C^\perp$ is an ideal linear code realizing $\Gamma_{\min}^*$, and its generator matrix is H. By the definition of an ideal linear code, there exist nonzero elements $g_{ij} \in \mathbb{F}_q$ that yield the matrix G constructed as above, and this matrix G serves as the parity-check matrix of $C^\perp$. Therefore, $GH^\mathsf{T} = 0$ holds over $\mathbb{F}_q$. □

Since the duality between an access structure and its dual is symmetric, the conditions for the existence of ideal linear codes realizing them also exhibit symmetry. Thus, we obtain the following theorem, which is consistent in formulation with Theorem 1 and essentially identical in nature.

Theorem 2 *If $\Gamma_{\min}^*$ can be realized by an ideal linear code D with parity-check matrix G, then $GH^\mathsf{T} = 0$ has a solution over $\mathbb{F}_q$.*

Proof The proof is similar to that of Theorem 1 and is therefore omitted. □

3.2 From Algebraic Conditions to the Construction of Schemes

In Section 3.1, we proved that if an access structure $\Gamma_{\min}$ is realized by an ideal linear code, then the system of quadratic equations $GH^{\mathsf{T}} = 0$ has a solution over $\mathbb{F}_q$. We now turn to the converse problem: assuming that the matrices H and G are constructed as in Section 3.1 and that $GH^{\mathsf{T}} = 0$ has a solution over $\mathbb{F}_q$, how can we construct a concrete ideal linear code that realizes the given access structure? That is, we seek an ideal linear code C with parity-check matrix H that realizes $\Gamma_{\min}$.

Theorem 3 *If $GH^{\mathsf{T}} = 0$ has a solution over $\mathbb{F}_q$, then $\Gamma_{\min}$ is realized by an ideal linear code C with parity-check matrix H.*

Proof Let C be the linear code with parity-check matrix H, and D be the linear code with parity-check matrix G. From the solvability of $GH^{\mathsf{T}} = 0$ over $\mathbb{F}_q$, we have $C^{\perp} \subseteq D$ and $D^{\perp} \subseteq C$.

Denote by $\Gamma_{\min}(C)$ the collection of minimal authorized sets corresponding to all minimal codewords with first component 1 in the row space of H; we call $\Gamma_{\min}(C)$ **the access structure induced by** C. Let $G^C = (\boldsymbol{g}_0, \boldsymbol{g}_1, \ldots, \boldsymbol{g}_n)$ be a generator matrix of C, with the labeling map ρ defined as

$$\rho = \begin{pmatrix} 2 & 3 & \cdots & n+1 \\ P_1 & P_2 & \cdots & P_n \end{pmatrix},$$

where the first row lists the column indices of G^C and the second row indicates the corresponding participants.

Since $D^{\perp} \subseteq C$, we necessarily have $\mathrm{span}\{G\} \subseteq \mathrm{span}\{G^C\} = C$. It remains to prove that $\Gamma_{\min}(C) = \Gamma_{\min}$, which then shows that the linear code C with parity-check matrix H is an ideal linear code realizing $\Gamma_{\min}$.

Each $S_i \in \Gamma_{\min}$ corresponds to the i-th row of H, denoted by $\boldsymbol{h}_i$. Because $G^C \boldsymbol{h}_i^\mathsf{T} = 0$, we have

$$\boldsymbol{g}_0 + h_{i1}\boldsymbol{g}_1 + \cdots + h_{in}\boldsymbol{g}_n = 0,$$

where $h_{ij} \neq 0$ if and only if $P_j \in S_i$. Hence S_i is an authorized set induced by C.

Suppose, for contradiction, that $S_i \notin \Gamma_{\min}(C)$, i.e., S_i is not a minimal authorized set induced by C. However, since S_i is authorized in $\Gamma_{\min}(C)$, it must contain some minimal authorized set $A \in \Gamma_{\min}(C)$ with $A \subsetneq S_i$. Because $S_i \in \Gamma_{\min}$ and any proper subset of a minimal authorized set is unauthorized, A is unauthorized with respect to $\Gamma_{\min}$. Therefore A is contained in some maximal unauthorized set corresponding to $\Gamma_{\min}$, say R_j, i.e., $A \subseteq R_j$.

By the construction of G, the set $S_j^* = P - R_j$ is a minimal authorized set of the dual access structure $\Gamma_{\min}^*$, and hence appears as some row $\boldsymbol{g}_j$ of G. It is easy to see that the first component of $\boldsymbol{g}_j$ is 1, the components corresponding to $P - R_j$ are nonzero, and those corresponding to R_j are zero (in particular, the positions corresponding to $A \subseteq R_j$ are zero). Without loss of generality, we may write $\boldsymbol{g}_j$ as

$$\boldsymbol{g}_j = (1, g_{j2}, \ldots, g_{jk}, 0, \ldots, 0).$$

Since the minimal authorized set $A \in \Gamma_{\min}(C)$, we may assume, without loss of generality, that the minimal codeword with first component 1 in the dual code of C corresponding to A is

$$\boldsymbol{c} = (1, 0, \ldots, 0, c_i, \ldots, c_n), \quad k < i,$$

so that $\boldsymbol{g}_j \boldsymbol{c}^\mathsf{T} = 1$.

But from $GH^\mathsf{T} = 0$, we have $\boldsymbol{g}_j \boldsymbol{c}^\mathsf{T} = 0$, a contradiction. Hence the assumption $S_i \notin \Gamma_{\min}(C)$ is false, and consequently $S_i \in \Gamma_{\min}(C)$.

Since S_i was arbitrary, we obtain $\Gamma_{\min} \subseteq \Gamma_{\min}(C)$. Now suppose there exists a minimal authorized set $S \in \Gamma_{\min}(C)$ such that $S \notin \Gamma_{\min}$. Then S is a minimal authorized set induced by C but is not a minimal authorized set of $\Gamma_{\min}$. In particular, S cannot be authorized with respect to $\Gamma_{\min}$; that is, S cannot contain any minimal authorized set $S_i \in \Gamma_{\min}$, for otherwise S and S_i would both belong to $\Gamma_{\min}(C)$, contradicting the minimality of S in $\Gamma_{\min}(C)$. Therefore S is unauthorized with respect to $\Gamma_{\min}$. By the same reasoning as above, we again derive a contradiction. Hence no such S exists, and we conclude $\Gamma_{\min} = \Gamma_{\min}(C)$.

By the labeling map ρ, the linear code C with parity-check matrix H is an ideal linear code realizing Γ_{min}. $\square$

From the constructions of H and G in Section 3.1, each row of H corresponds to a minimal authorized set of Γ_{min}, while each row of G corresponds to a minimal authorized set of the dual access structure Γ^*_{min}. Thus we obtain the following theorem, which is symmetric to Theorem 3.

Theorem 4 *If $GH^\top = 0$ has a solution over $\mathbb{F}_q$, then Γ^*_{min} is realized by an ideal linear code D with parity-check matrix G.*

Proof The proof is similar to that of Theorem 3 and is therefore omitted. $\square$

By Theorem 3, any solution H of $GH^\top = 0$ over $\mathbb{F}_q$ is precisely the parity-check matrix of an ideal linear code C realizing Γ_{min}. Similarly, Theorem 4 shows that the solution matrix G simultaneously serves as the parity-check matrix of an ideal linear code D realizing the dual access structure Γ^*_{min}.

3.3 Corollary and Establishment of the Necessary and Sufficient Theorem

From Section 3.2, a solution to $GH^\top = 0$ over $\mathbb{F}_q$ not only provides an ideal linear secret sharing scheme for Γ_{min}, but also provides an ideal linear secret sharing scheme for Γ^*_{min}. We now present a corollary (see corollary 6) from the perspective of matroid theory, which shows that provided that $GH^\top = 0$ has a solution over $\mathbb{F}_q$ under the matrix constructions given in Section 3.1, it follows that the ideal linear codes which realize two mutually dual access structures are themselves dual to one another.

Before proceeding to the proof of the corollary, we first state and confirm a key foundational result, namely the following lemma 5.

Lemma 5 ([22, Theorem 1]) *Let $\Gamma_{\min}$ be a non-redundant access structure defined on a participant set P. If $\Gamma_{\min}$ is ideal, then there exists a connected matroid $\mathcal{M}$ on the finite set $P \cup \{P_0\}$ such that $\Gamma_{\min}$ is the port of $\mathcal{M}$ at P_0.*

Remark 3 All ports of a connected matroid are non-redundant access structures. Moreover, if a non-redundant access structure is the port of a matroid, then that matroid is necessarily connected.

By [30, Theorem 4.3.2] , such a connected matroid is unique, and it is unique up to isomorphism. This fact is widely accepted by subsequent researchers; see, for example, Hameed [32, pp. 135].

Corollary 6 *If the system of equations $GH^\mathsf{T} = 0$ has a solution over $\mathbb{F}_q$, then* $\mathrm{rank}_{\mathrm{LIN}}(G) + \mathrm{rank}_{\mathrm{LIN}}(H) = n + 1$.

Proof Since $GH^\mathsf{T} = 0$ has a solution over $\mathbb{F}_q$, the access structure $\Gamma_{\min}$ is realized by an ideal linear code C with parity-check matrix H; let $\mathcal{M}_C$ denote the representable connected matroid associated with this code. Simultaneously, the dual access structure $\Gamma^*_{\min}$ is realized by an ideal linear code D with parity-check matrix G, and let $\mathcal{M}_D$ denote the representable connected matroid associated with D.

Furthermore, $\Gamma^*_{\min}$ is also realized by the ideal linear code $C^\perp$ with generator matrix H, and the representable connected matroid associated with this code, denoted by $\mathcal{M}_{C^\perp}$, is the dual matroid of $\mathcal{M}_C$.

Since $\Gamma^*_{\min}$ is realized by two ideal linear codes, namely D and $C^\perp$, it follows from Lemma 5 that such a matroid is unique up to isomorphism; i.e., $\mathcal{M}_{C^\perp} \cong \mathcal{M}_D$. Consequently, their ranks (and hence the ranks of their corresponding representing matrices) are equal. Let H_D be a generator matrix of D. Then $\mathrm{rank}_{\mathrm{LIN}}(H) = \mathrm{rank}_{\mathrm{LIN}}(H_D) = k$. Since G is the parity-check matrix of D, we have $\mathrm{rank}_{\mathrm{LIN}}(G) = n + 1 - k$. Therefore,

$$\mathrm{rank}_{\mathrm{LIN}}(G) + \mathrm{rank}_{\mathrm{LIN}}(H) = n + 1.$$

$\square$

By Theorems 3 and 4, if $GH^\top = 0$ has a solution over $\mathbb{F}_q$, then $\Gamma_{\min}$ is realized by an ideal linear code C with parity-check matrix H, and $\Gamma^*_{\min}$ is realized by an ideal linear code D with parity-check matrix G. The corollary further shows that D is exactly the dual code of C, i.e., $D = C^\perp$. Indeed, from $GH^\top = 0$ we have

$$D = \{\boldsymbol{v} \in \mathbb{F}_q^{n+1} \mid G\boldsymbol{v}^\top = 0\}, \qquad C = \{\boldsymbol{c} \in \mathbb{F}_q^{n+1} \mid H\boldsymbol{c}^\top = 0\}.$$

It is easy to see that $C^\perp \subseteq D$. The corollary gives $\mathrm{rank}_{\mathrm{LIN}}(G) + \mathrm{rank}_{\mathrm{LIN}}(H) = n+1$, hence

$$\dim(D) = n + 1 - \mathrm{rank}_{\mathrm{LIN}}(G) = \mathrm{rank}_{\mathrm{LIN}}(H) = \dim(C^\perp),$$

so $C^\perp = D$.

Based on the constructions of G and H given in Section 3.1, together with Theorems 1–4 and Corollary 6 above, we now establish the following core theorem.

Theorem 7 *Let $P = \{P_1, \dots, P_n\}$ be a set of n participants, and $\Gamma_{\min} = \{S_1, \dots, S_m\}$ be an access structure on P with dual access structure $\Gamma^*_{\min} = \{S^*_1, \dots, S^*_l\}$. Let $\mathbb{F}_q$ be a finite field. Then the following three statements are equivalent:*

1. *The system of equations $GH^\top = 0$ has a solution over $\mathbb{F}_q$;*
2. *The access structure $\Gamma_{\min}$ is realized by an ideal linear code C over $\mathbb{F}_q$, with G serving as a generator matrix and H as its parity-check matrix;*
3. *The dual access structure $\Gamma^*_{\min}$ is realized by an ideal linear code over $\mathbb{F}_q$, which is precisely the dual code $C^\perp$ of C, with H as its generator matrix and G as its parity-check matrix.*

From Theorem 7, determining whether an access structure can be realized by an ideal linear secret sharing scheme is equivalent to determining whether the system of equations $GH^{\top} = 0$ associated with a specifically constructed linear code has a solution over $\mathbb{F}_q$. This theorem transforms the abstract problem of the existence of an ideal linear secret sharing scheme into a concrete and solvable algebraic problem. Indeed, Karchmer et al. [7] showed that for every access structure there exists a linear secret sharing scheme realizing it, but whether an arbitrary access structure can be realized by an ideal linear secret sharing scheme has remained unknown. Theorem 7 thus resolves this question. Moreover, whether the dual access structure corresponding to the given access structure can be realized by an ideal linear secret sharing scheme is also equivalent to the same condition.

4 Unified Framework and Illustrative Examples

4.1 Construction of a Unified Framework

Brickell and Davenport [22] provided a partial characterization of ideal access structures, namely, if an access structure is the port of a matroid representable over some finite field, then the access structure is realized by an ideal linear secret sharing scheme over that field (sufficient condition); if an access structure is ideal, then it is the port of some (unique up to isomorphism) matroid (necessary condition). These sufficient and necessary conditions do not fully coincide; there exists a gap. For example, Seymour [33] proved that the access structure induced by the Vámos matroid is not ideal, demonstrating that the necessary condition is not sufficient.

In fact, the converse of the sufficient condition given by Brickell and Davenport also holds: if an access structure can be realized by an ideal linear secret sharing scheme over a finite field $\mathbb{F}_q$, then the matrix corresponding to this scheme defines an $\mathbb{F}_q$-representable matroid, and consequently the access structure is the port of this representable matroid over that field. Moreover, monotone span programs are equivalent to linear secret sharing schemes [3, 7], and there exists a one-to-one correspondence between linear codes and monotone span programs [9]. Therefore, we can prove this converse statement from the perspective of ideal linear codes (see Theorem 8).

Although the connection between an ideal linear secret sharing scheme realizing an access structure and a corresponding representable matroid was already implicit from the perspective of linear codes and matroids in Section 2.3, we still provide an independent proof of this statement (see Theorem 8), mainly in view of the unified framework we are about to establish. Before giving the proof, we present an alternative definition of a matroid, different from Definition 5.

Definition 10 (Matroid (rank function definition)) A matroid $\mathcal{M} = (V, r_{\mathcal{M}})$ consists of a finite set V and a function $r_{\mathcal{M}} : \mathcal{P}(V) \to \mathbb{N}$ satisfying the following three axioms:

- **Non-negativity and Boundedness:** For any $X \subseteq V$, $0 \leq r_{\mathcal{M}}(X) \leq |X|$.
- **Monotonicity:** For any $X \subseteq Y \subseteq V$, $r_{\mathcal{M}}(X) \leq r_{\mathcal{M}}(Y)$.
- **Submodularity:** $r_{\mathcal{M}}(X \cup Y) + r_{\mathcal{M}}(X \cap Y) \leq r_{\mathcal{M}}(X) + r_{\mathcal{M}}(Y)$.

In a matroid $\mathcal{M}$, a subset $X \subseteq V$ is **independent** if $r_{\mathcal{M}}(X) = |X|$; it is **dependent** if $r_{\mathcal{M}}(X) < |X|$. A subset $A \subseteq V$ is a **circuit** if it is a minimal dependent set (i.e., for every $a \in A$, $r_{\mathcal{M}}(A) = |A| - 1$ and $r_{\mathcal{M}}(A - \{a\}) = |A| - 1$).

Let C be an ideal linear code over $\mathbb{F}_q$ realizing the access structure $\Gamma_{\min}$, with $k \times (n+1)$ generator matrix

$$G = (\boldsymbol{g}_0, \boldsymbol{g}_1, \ldots, \boldsymbol{g}_n),$$

and the corresponding labeling map ρ given by

$$\rho = \begin{pmatrix} 2 & 3 & \cdots & n+1 \\ P_1 & P_2 & \cdots & P_n \end{pmatrix},$$

where the first row lists the column indices of G and the second row indicates the corresponding participants.

Define the finite set $V = \{P_0, P_1, \ldots, P_n\}$ and the mapping $\phi : V \to \mathbb{F}_q^k$ by $\phi(P_i) = \boldsymbol{g}_i$. For any $A \subseteq V$, define

$$r_{\mathcal{M}}(A) = \mathrm{rank}_{\mathrm{LIN}}\big(\{\phi(P_i) \mid P_i \in A\}\big).$$

Then the following theorem holds.

Theorem 8 *Let $\Gamma_{\min}$ be an access structure defined on the participant set $P = \{P_1, \ldots, P_n\}$ that is realized by an ideal linear code C over $\mathbb{F}_q$. Then $\Gamma_{\min}$ is the port at P_0 of an $\mathbb{F}_q$-representable matroid $\mathcal{M} = (V, r_{\mathcal{M}})$.*

Proof The function $r_{\mathcal{M}}$ clearly satisfies non-negativity, boundedness, and monotonicity. Moreover, for any $A, B \subseteq V$, we have

$$\begin{aligned} r_{\mathcal{M}}(A \cup B) + r_{\mathcal{M}}(A \cap B) &\le r_{\mathcal{M}}(\mathrm{span}(A) \cup \mathrm{span}(B)) + r_{\mathcal{M}}(\mathrm{span}(A) \cap \mathrm{span}(B)) \\ &= r_{\mathcal{M}}(A) + r_{\mathcal{M}}(B), \end{aligned}$$

so $r_{\mathcal{M}}$ satisfies submodularity. Hence $\mathcal{M} = (V, r_{\mathcal{M}})$ is a matroid.

By construction, the mapping $\phi : V \to \mathbb{F}_q^k$ shows that $\mathcal{M}$ is representable over $\mathbb{F}_q$.

Without loss of generality, assume that $S = \{P_1, P_2, \ldots, P_l\}$ is a minimal authorized set in $\Gamma_{\min}$. Then there exist nonzero coefficients $\alpha_{P_i} \in \mathbb{F}_q$ such that

$$\boldsymbol{g}_0 = \sum_{P_i \in S} \alpha_{P_i} \boldsymbol{g}_i,$$

thus $r_{\mathcal{M}}(S \cup \{P_0\}) < l + 1$, i.e., $S \cup \{P_0\}$ is a dependent set. For any $P_i \in S \cup \{P_0\}$:

1. Case 1: $P_i = P_0$. Then $r_{\mathcal{M}}(S) = l$ (otherwise this would contradict the minimality of S), so S is independent.

2. Case 2: $P_i \in S$. Let $S' = S - \{P_i\}$. Then $\boldsymbol{g}_0$ cannot be linearly expressed by S'; hence

$$r_{\mathcal{M}}(S' \cup \{P_0\}) = 1 + r_{\mathcal{M}}(S').$$

From Case 1, S' is independent (as a subset of an independent set), so $r_{\mathcal{M}}(S') = l - 1$. Therefore $r_{\mathcal{M}}(S' \cup \{P_0\}) = l$, i.e., $S' \cup \{P_0\}$ is independent.

Now suppose $S \cup \{P_0\}$ is a circuit of $\mathcal{M}$. By the definition of a circuit, we have $r_{\mathcal{M}}(S \cup \{P_0\}) = l$ and $r_{\mathcal{M}}(S) = l$. Hence there exist nonzero coefficients such that $\boldsymbol{g}_0$ can be linearly expressed by $\{\boldsymbol{g}_i \mid P_i \in S\}$; therefore S is an authorized set. If S were not minimal, then it would contain a minimal authorized set $S' \subsetneq S$. Then $\boldsymbol{g}_0 \cup \{\boldsymbol{g}_i \mid P_i \in S'\}$ would be linearly dependent, implying $r_{\mathcal{M}}(S' \cup \{P_0\}) < |S'| + 1$, which contradicts the fact that $S \cup \{P_0\}$ is a circuit of $\mathcal{M}$. Hence S is a minimal authorized set. □

It is worth noting that, from the above proof of the converse of Brickell and Davenport's sufficient condition based on the perspective of ideal linear codes, it follows that the statement "an access structure is realized by an ideal linear secret sharing scheme over some finite field if and only if the access structure is the port of a matroid representable over that field" holds true. When we revisit those families of access structures for which the ideal access structures have already been characterized, we find that these specific families satisfy the above proposition. For example, Tassa [24] constructed ideal linear secret sharing schemes for hierarchical threshold access structures using Birkhoff interpolation; similar results exist for the families of graph-defined access structures [22, 25, 26], bipartite access structures [27], and access structures with three or four minimal authorized sets [28].

Although this equivalence has been established for specific families of access structures, it does not resolve the problem of a universal criterion for arbitrary access structures, nor does it provide a general method of verification. Therefore, from an algebraic perspective, we have given necessary and sufficient conditions for the existence of an ideal linear secret sharing scheme for an arbitrary access structure. This condition states that an arbitrary access structure can be realized by an ideal linear secret sharing scheme over some finite field if and only if the system of equations $GH^{\mathsf{T}} = 0$ has a solution over that field.

Moreover, our necessary and sufficient condition unifies the descriptions of the same mathematical object from three different fields: cryptography (secret sharing), algebraic coding theory (linear codes), and combinatorics (representable matroids). Namely,

$$\begin{aligned}
&\text{An access structure is realized by an ideal linear secret sharing scheme over } \mathbb{F}_q \\
&\qquad \iff GH^{\mathsf{T}} = 0 \text{ has a solution over } \mathbb{F}_q \\
&\qquad \iff \text{The access structure is the port of an } \mathbb{F}_q\text{-representable matroid.}
\end{aligned}$$

The significance of this unified framework lies in the fact that, theoretically, analyzing whether an access structure (or a family thereof) can be realized by an ideal linear secret sharing scheme is equivalent to studying whether its corresponding matroid is representable over some field (e.g., the (t, n)-threshold access structure corresponds to the uniform matroid $U_{t,n+1}$, and $U_{t,n+1}$ is representable over $\mathbb{F}_q$ whenever $q \geq n$([30, p. 202]), and the existence of an ideal linear secret sharing scheme is also equivalent to solving $GH^{\mathsf{T}} = 0$ over $\mathbb{F}_q$. This means that we can study the same mathematical object using different tools from different perspectives, thereby providing a computable and verifiable practical tool for systematically addressing the problem of the existence of ideal linear secret sharing schemes for arbitrary access structures.

Since the necessary and sufficient conditions proposed in this paper are intended to determine whether an access structure can be realized by an ideal linear secret sharing scheme, it follows that for the access structure induced by the Vámos matroid, one can theoretically verify that the matrices G and H constructed from this access structure yield no solution to $GH^{\mathsf{T}} = 0$ over any field. This implies that the access structure induced by the Vámos matroid cannot be realized by an ideal linear secret sharing scheme; however, whether it can be realized by an ideal nonlinear secret sharing scheme remains unknown. Therefore, the theorem presented in this paper does not address the gap between the sufficient and necessary conditions given by Brickell and Davenport [22]; it can only be used to determine, in theory, whether an access structure can be realized by an ideal linear secret sharing scheme, but cannot determine whether it can be realized by an ideal nonlinear one.

4.2 Application and Verification of the Unified Framework

In Section 4.1, we established a unified framework based on whether the system of equations $GH^{\mathsf{T}} = 0$ has a solution over a finite field. This framework theoretically reveals that determining whether an arbitrary access structure can be realized by an ideal linear secret sharing scheme over some finite field is equivalent to solving a concrete algebraic problem over that field, and also equivalent to finding a representable matroid over that field. Therefore, this subsection selects an access structure as an example to verify the unified framework, thereby providing a clear paradigm for its application to other access structures.

Example 1 Let the set of participants be $P = \{P_1, P_2, P_3, P_4, P_5, P_6\}$, and let the access structure defined on P be

$$\Gamma_{\min} = \big\{\{P_1, P_2, P_4, P_5\},\ \{P_1, P_2, P_4, P_6\},\ \{P_1, P_3, P_4, P_5\},\ \{P_1, P_3, P_4, P_6\}\big\}.$$

We now determine whether there exists an ideal linear code over the finite field $\mathbb{F}_{11}$ that realizes $\Gamma_{\min}$.

The dual access structure corresponding to $\Gamma_{\min}$ is

$$\Gamma^*_{\min} = \big\{\{P_5, P_6\},\ \{P_4\},\ \{P_2, P_3\},\ \{P_1\}\big\}.$$

Thus, following the construction in Section 3.1, we construct the matrices H and G over $\mathbb{F}_{11}$ as follows:

$$H = \begin{pmatrix} 1 & h_{11} & h_{12} & 0 & h_{14} & h_{15} & 0 \\ 1 & h_{21} & h_{22} & 0 & h_{24} & 0 & h_{26} \\ 1 & h_{31} & 0 & h_{33} & h_{34} & h_{35} & 0 \\ 1 & h_{41} & 0 & h_{43} & h_{44} & 0 & h_{46} \end{pmatrix}, \qquad G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & g_{15} & g_{16} \\ 1 & 0 & 0 & 0 & g_{24} & 0 & 0 \\ 1 & 0 & g_{32} & g_{33} & 0 & 0 & 0 \\ 1 & g_{41} & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where $h_{ij} \neq 0$ if and only if $P_j \in S_i$ (each row of H corresponds to a minimal authorized set $S_i \in \Gamma_{\min}$), and $g_{ij} \neq 0$ if and only if $P_j \in S_i^*$ (each row of G corresponds to a minimal authorized set $S_i^* \in \Gamma_{\min}^*$). The labeling map for the columns of H and G follows that given in Section 3.1.

Solving $GH^{\mathsf{T}} = 0$ over $\mathbb{F}_{11}$ is equivalent to solving the following system of equations:

$$\begin{cases} 1 + h_{15}g_{15} = 0, \\ 1 + h_{14}g_{24} = 0, \\ 1 + h_{12}g_{32} = 0, \\ 1 + h_{11}g_{41} = 0, \\ 1 + h_{26}g_{16} = 0, \\ 1 + h_{24}g_{24} = 0, \\ 1 + h_{22}g_{32} = 0, \\ 1 + h_{21}g_{41} = 0, \\ 1 + h_{35}g_{15} = 0, \\ 1 + h_{34}g_{24} = 0, \\ 1 + h_{33}g_{33} = 0, \\ 1 + h_{31}g_{41} = 0, \\ 1 + h_{46}g_{16} = 0, \\ 1 + h_{44}g_{24} = 0, \\ 1 + h_{43}g_{33} = 0, \\ 1 + h_{41}g_{41} = 0. \end{cases}$$

This system has a general solution:

$$\begin{aligned} &h_{15} = g_{15}^{-1},\ h_{14} = g_{24}^{-1},\ h_{12} = g_{32}^{-1},\ h_{11} = g_{41}^{-1}, \\ &h_{26} = g_{16}^{-1},\ h_{24} = g_{24}^{-1},\ h_{22} = g_{32}^{-1},\ h_{21} = g_{41}^{-1}, \\ &h_{35} = g_{15}^{-1},\ h_{34} = g_{24}^{-1},\ h_{33} = g_{33}^{-1},\ h_{31} = g_{41}^{-1}, \\ &h_{46} = g_{16}^{-1},\ h_{44} = g_{24}^{-1},\ h_{43} = g_{33}^{-1},\ h_{41} = g_{41}^{-1}. \end{aligned}$$

Therefore, there exists a set of solutions over $\mathbb{F}_{11}$ such that the ideal linear code C with parity-check matrix H realizes $\Gamma_{\min}$; the generator matrix of this linear code is exactly G.

Since the duality between access structures is mutual, under the same set of solutions, the ideal linear code $D = C^{\perp}$ with parity-check matrix G simultaneously realizes $\Gamma^*_{\min}$; this code is precisely the dual code of C, and H serves as its generator matrix.

Furthermore, the matrices constructed above directly yield an $\mathbb{F}_{11}$-representable matroid $\mathcal{M}$ associated with the access structure $\Gamma_{\min}$. Suppose the parity-check matrix H of the ideal linear code C has rank k; then its generator matrix G has rank $n+1-k$. As described in Section 4.1, write the generator matrix of C as $G = (\boldsymbol{g}_0, \boldsymbol{g}_1, \ldots, \boldsymbol{g}_n)$. Let $V = \{P_0, P_1, P_2, P_3, P_4, P_5, P_6\}$ and define a mapping $\phi : V \to \mathbb{F}_{11}^{n+1-k}$ by $\phi(P_i) = \boldsymbol{g}_i$. Further define $r_{\mathcal{M}}(A) = \mathrm{rank}_{\mathrm{LIN}}\left(\{\phi(P_i) \mid P_i \in A,\ A \subseteq V\}\right)$. Then $\Gamma_{\min}$ is the port of the matroid $\mathcal{M} = (V, r_{\mathcal{M}})$ at P_0.

Similarly, let $H = (\boldsymbol{h}_0, \boldsymbol{h}_1, \ldots, \boldsymbol{h}_n)$ be the generator matrix of the ideal linear code D realizing the dual access structure $\Gamma^*_{\min}$. Then over $\mathbb{F}_{11}$, $\Gamma^*_{\min}$ is the port at P_0 of the representable matroid $\mathcal{M}^* = (V, r_{\mathcal{M}^*})$, where $\phi^* : V \to \mathbb{F}_{11}^{k}$ is defined by $\phi^*(P_i) = \boldsymbol{h}_i$.

5 Conclusion

In this paper, necessary and sufficient conditions are established for the existence of an ideal linear secret sharing scheme for an arbitrary access structure. The problem is reduced to the solvability of the system of equations $GH^{\mathsf{T}} = 0$ associated with a linear code of a specific construction over a finite field, thereby providing a universal algebraic criterion. This condition simultaneously reveals an essential equivalence among cryptography, algebraic coding theory, and combinatorics in characterizing the same mathematical object. However, this criterion is applicable only to determining whether a given access structure can be realized by an ideal linear secret sharing scheme; it does not address the gap in the sufficient and necessary conditions given by Brickell and Davenport [22] concerning access structures that are ports of a matroid but are not ideal (e.g., the access structure induced by the Vámos matroid), and thus cannot determine whether such structures can be realized by ideal nonlinear schemes. Finally, the proposed conditions do not provide a method for selecting an appropriate finite field $\mathbb{F}_q$. Consequently, the absence of a solution over a given field does not preclude the existence of a solution over another field, leading to the following open problem:

Open Problem. Given an access structure, how can one characterize the existence and distribution of solutions to the system $GH^\top = 0$ over different finite fields?

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